

PERSONAL KNOWLEDGE BASE: YASHWANTH KUMAR KOTLA

DOCUMENT PURPOSE

This document is a structured knowledge base about Yashwanth Kumar Kotla, intended as source material for a personal RAG (Retrieval-Augmented Generation) chatbot. Each section is self-contained so it can be chunked and retrieved independently. Section headers describe the topic of the section.

SECTION: IDENTITY AND CONTACT

Full name: Yashwanth Kumar Kotla.

Location: Austin, Texas, USA.

Email: yashwanthkumarkotla@gmail.com.

Phone: +1 (804) 486-0861.

LinkedIn: linkedin.com/in/yashwanthkumarkotla.

Visa status: F-1 visa, with 3-year STEM OPT eligibility after graduation.

Professional identity: Data Scientist and AI Product Developer with hands-on experience building and deploying end-to-end machine learning systems, predictive analytics platforms, AI applications, and scalable data pipelines.

SECTION: EDUCATION

Degree in progress: Master of Science in Data Science, Webster University, Austin, Texas.

Program dates: August 2024 to December 2026.

Status as of mid-2026: Two subjects remaining before graduation. Graduating December 2026.

Background degree: Bachelor of Technology (B.Tech) in Information Technology, providing a foundation in data structures, algorithms (DSA), and Python.

Certifications: Google Data Analytics Professional Certificate. Oracle Cloud Infrastructure (OCI) Foundations Associate.

Skill gaps identified entering the MS program: statistics depth, deep learning, and MLOps. Core baseline skills entering the program: Pandas, NumPy, Matplotlib, scikit-learn, and basic supervised machine learning.

SECTION: PUBLICATION

Title: "High Dimensional Text Classification Using Unsupervised Machine Learning Algorithm."

Venue: IEEE ICAAI 2024 (International Conference on Advances in Artificial intelligence and machine learning), pages 368 to 372.

DOI: 10.1109/ICAII60222.2024.10575444.

Summary: Proposed enhancements to Naive Bayes classification and K-means clustering for dimensionality reduction in high-dimensional text data.

Note: This paper was verified as legitimate and is presented on the resume and LinkedIn using "co-authored" framing, anchored with the DOI.

SECTION: CAREER GOALS AND JOB SEARCH STRATEGY

Target roles: Data Scientist and MLOps Engineer.

Current phase: Yashwanth has been in a deliberate skill-building phase, intentionally not yet applying broadly to jobs. He has stated he will tell his AI assistant when he is ready to begin active applications, except for select opportunities he has already pursued (see Job Search Activity section).

Eligibility: 3-year STEM OPT eligibility under F-1 visa status after graduating in December 2026.

Long-term plan: A multi-month roadmap was built spanning skill phases: production ML and deployment fundamentals, NLP and LLM engineering (including RAG), MLOps and cloud (AWS) basics, and portfolio and interview preparation.

SECTION: JOB SEARCH ACTIVITY

P&G (Procter & Gamble): Applied for a Data Scientist role. Progressed through Aon's G.A.T.E. gamified assessment (covering Switch, Grid, and Digit challenges) and the PEAK behavioral assessment. Experienced a technical issue during the process when a server maintenance outage prevented a submission from registering correctly.

Recruiter outreach declined: Outreach from a staffing mill (PCS/CCS Global Tech) was received and declined. Outreach from Apetan Consulting for an ML contract role was also declined, after being advised to frame the experience gap honestly if pursued.

Alignerr: Researched as a potential gig annotation platform (run by Labelbox) for supplemental income.

Brain Co. cold outreach: Conducted targeted cold outreach strategy toward Brain Co., including researching co-founders and drafting emails tailored to different team members. The outreach was anchored on the Churn Intelligence System and Aura Duo projects, supported by a blind pairwise evaluation harness (aura_duo_eval.py) built specifically to generate real, defensible performance metrics for the outreach.

SECTION: CORE PORTFOLIO PROJECT - AURA DUO

Name: Aura Duo.

Role: Founder / AI Product Developer.

Duration: September 2025 to March 2026.

Live at: auraduo.app.

Description: An AI-powered dual-intelligence platform that orchestrates GPT-4 and Gemini to collaboratively generate, critique, and synthesize responses for higher-quality reasoning and reduced hallucinations.

Architecture: A multi-agent LLM architecture in which independent model outputs undergo cross-evaluation, critique scoring, and response refinement before producing a final synthesized answer, via a four-stage critique loop.

Engineering details: Built prompt orchestration workflows, evaluation pipelines, and model interaction frameworks. Integrated OpenAI and Google Gemini APIs into a scalable production application, handling model routing, response aggregation, error handling, and user interaction workflows.

Evaluation: A blind pairwise evaluation harness (`aura_duo_eval.py`) was built to generate real performance metrics for use in job outreach and portfolio presentation.

Significance: Identified as Yashwanth's strongest portfolio differentiator during a full GitHub audit, and used as the anchor project for high-stakes applications, including outreach toward Neuralink and Brain Co.

SECTION: CORE PORTFOLIO PROJECT - LAYOFF RADAR

Name: Layoff Radar (also referred to as the Tech Layoff Early Warning System), described on the resume as "Predictive Risk Intelligence Platform."

Description: An end-to-end workforce risk intelligence platform that predicts the likelihood of corporate layoffs using financial performance indicators, historical layoff events, and executive sentiment signals extracted from earnings communications.

Data: Integrates 47,000+ quarterly financial records from SimFin and 2,300+ historical layoff events (originally sourced including `layoffs.fyi` and `yfinance`), using a temporal labeling framework to turn historical events into predictive training examples for early-warning risk detection.

Modeling: XGBoost-based predictive models optimized using SMOTE for class imbalance, hyperparameter tuning, probability calibration, and business-driven evaluation metrics.

NLP component: A financial NLP pipeline using FinBERT automatically retrieves and analyzes CEO and CFO earnings-call transcripts, extracting sentiment and management-confidence signals used as predictive features.

Explainability: SHAP-based explainability provides transparent feature attribution so users can understand the financial, operational, and sentiment factors behind individual layoff-risk predictions.

Stack: FastAPI for REST APIs, Docker for containerization, MLflow for experiment tracking, supporting both single-company and batch risk assessments. Status: backtesting was pending as of the most recent update.

Status: Live app, with code on GitHub.

SECTION: CORE PORTFOLIO PROJECT - CHURN INTELLIGENCE SYSTEM (RETENTION ROI SYSTEM)

Name: Churn Intelligence System, also described on the resume as "Retention ROI System."

Description: A multi-layer retention analytics system integrating predictive modeling, causal inference, explainable AI, and LLM-driven decision support to quantify churn risk, intervention effectiveness, and expected retention ROI.

Modeling: End-to-end ML pipelines using Scikit-learn, XGBoost, and LightGBM, with calibrated probability estimation, SMOTE for class-imbalance handling, AUC-PR optimization, and business-driven, cost-weighted threshold optimization.

Uplift modeling: A CausalML uplift modeling layer (planned/in development) estimates treatment effects and identifies customer segments most likely to respond positively to retention interventions, improving targeting efficiency beyond standard churn prediction.

Explainability and narrative generation: SHAP explanations are combined with LLM-based narrative synthesis to automatically generate customer-specific, human-readable retention strategy recommendations for customer success and marketing teams.

Deployment: Containerized REST APIs and interactive business dashboards, with model monitoring, drift detection, and experiment tracking for production-grade MLOps.

Saved artifacts referenced in development: xgb_calibrated.pkl, scaler.pkl, threshold.txt.

SECTION: CORE PORTFOLIO PROJECT - LLM RESEARCH EVALUATION PLATFORM

Description: A multi-agent RAG evaluation platform that benchmarked 4 or more foundation models across 1,000+ enterprise knowledge retrieval tasks.

Framework: An automated experimentation framework using LangGraph and Ragas to measure hallucination rate, context relevance, retrieval precision, latency, and cost efficiency.

Retrieval: Hybrid retrieval pipelines combining semantic search, metadata filtering, and reranking, improving answer accuracy by 30% compared to baseline vector retrieval.

Output: Quantitative performance reports and comparative rankings across proprietary and open-source LLMs, plus an interactive Streamlit dashboard for experiment tracking, model comparison, and performance visualization.

Significance: Converted academic AI evaluation methodologies into production-ready experimentation tooling, reducing manual benchmarking effort by approximately 80%.

SECTION: CORE PORTFOLIO PROJECT - INSURANCE PREMIUM CATEGORY PREDICTOR (MEDICAL INSURANCE COST PREDICTOR)

Description: An end-to-end insurance risk classification platform that predicts premium categories using demographic, lifestyle, and behavioral risk factors.

Modeling: Scikit-learn pipelines using engineered features including BMI, age segmentation, lifestyle risk scoring, and city-tier categorization. Reported R-squared of 0.77 in earlier development.

Deployment: FastAPI and Pydantic for schema-validated RESTful inference APIs with health monitoring endpoints. A Streamlit-based decision-support interface allows interactive premium category predictions. Containerized with Docker and published to Docker Hub as [yashkotla/streamlit-insurance-cat-predictor](#).

Status: Live app, with code on GitHub. This was one of Yashwanth's early portfolio projects.

SECTION: ACTIVE PROJECT - MANA RAITHU (FARMING CALCULATOR APP)

Name: Mana Raithu.

Description: An offline-first Android farming calculator app targeting Indian farmers, including elderly and low-literacy users.

Tech stack: Flutter, targeting offline-first architecture with support for Telugu, Hindi, and English.

Phase 1 scope (deliberately limited): exactly four calculators — Land Area Converter, Crop Profit/Loss Estimator, Seed Rate Calculator, and Loan/EMI Calculator — plus history storage and localization scaffolding. Weather, mandi (market) prices, and AI features were explicitly deferred and are not part of Phase 1.

UX constraints driving design: flat navigation with no nested menus, icon-plus-label buttons, large touch targets (minimum 56dp), high contrast, numeric inputs preferred over text inputs, one task per screen, and always-visible home/back buttons, based on UX research for low-literacy and rural Indian users.

Product-market-fit assessment: A structured PMF validation (using a custom-built kill-funnel framework) found weak market signal — every Phase 1 calculator already exists as a free competing app or website (for example, [agrifarming.in](#) alone has 89+ calculators), well-funded incumbents exist (Mahindra-backed Krish-e, IFFCO, Krishi Network), and no clear monetization or willingness-to-pay signal was found. Yashwanth clarified that monetary return is not his primary motivation for this project, so development continued past the PMF concerns for learning and execution purposes.

Development environment: Built on an M1 MacBook using Android Studio and Flutter, targeting a Pixel 6 API 33 Android Virtual Device (AVD) with arm64-v8a system images (required for M1 architecture compatibility).

SECTION: OTHER PROJECTS

JobLens: A job search aggregator built in Python and Flask, using multi-source scraping with parallel execution via `ThreadPoolExecutor` and a single-page application (SPA) frontend. Went through production redesign and scraper debugging iterations.

LSTM Next-Word Predictor: A next-word prediction model trained on the Sherlock Holmes dataset, built and debugged in Google Colab then migrated to local deployment, eventually wrapped in a Streamlit interface as a small applied NLP project following LSTM coursework.

Walmart Time Series Forecasting: An academic assignment comparing ARIMA, SARIMA, SARIMAX, and Prophet forecasting models, delivered as a Word document plus a Jupyter notebook. Related Walmart exploratory data analysis (EDA) work was also completed.

Auto-Apply Pipeline (system design, conceptual): A designed (not necessarily fully built) daily job-application automation pipeline concept covering job discovery, classification, JD (job description) NLP analysis, a sponsorship gate, resume customization, cover letter generation, application-form intelligence, auto-apply versus manual review queuing, and tracking/analytics, intended to run on a daily cron schedule with a shared PostgreSQL database.

SECTION: WORK EXPERIENCE - FREELANCE DATA SCIENTIST

Role: Data Scientist (freelance/gig).

Location: Austin, Texas.

Duration: April 2025 to September 2025.

Responsibilities: Developed automated ETL pipelines integrating data from multiple business sources, improving data accessibility and reducing manual reporting effort by approximately 40%. Built interactive Tableau and Power BI dashboards to monitor operational KPIs, customer behavior, and campaign performance. Conducted exploratory data analysis, statistical testing, and business performance analysis to optimize marketing strategy and campaign ROI. Collaborated with stakeholders to translate business requirements into analytical solutions. Designed data models and reporting workflows to improve reporting accuracy and executive-level visibility.

SECTION: TECHNICAL SKILLS

Programming languages: Python, SQL.

Machine learning and AI: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, SHAP, CausalML, FinBERT, Transformers, OpenAI API, Google Gemini API, LangChain, deep learning, natural language processing (NLP), generative AI, large language models (LLMs), pretraining, supervised fine-tuning, post-training, alignment, model evaluation, scientific data handling.

Data science and analytics: statistical modeling, hypothesis testing, A/B testing, predictive analytics, classification, feature engineering, model calibration, uplift modeling, customer segmentation, explainable AI (XAI), exploratory data analysis (EDA).

Data engineering: ETL pipelines, data warehousing, data modeling, data cleaning, feature stores, data integration, PostgreSQL, MySQL.

MLOps and deployment: FastAPI, Streamlit, Docker, Docker Compose, MLflow, Git, GitHub Actions, REST APIs, model monitoring, drift detection, CI/CD.

Cloud and infrastructure: AWS (S3, EC2, SageMaker), Google Cloud Platform (GCP), Render.

Visualization and BI: Tableau, Power BI, Plotly, Matplotlib, Seaborn.

Developer tools: Git, GitHub, Jupyter Notebook, VS Code, Postman.

Current learning focus (as of mid-2026): RAG (Retrieval-Augmented Generation) architecture and implementation — including this very project, a personal RAG chatbot built on this knowledge base.

SECTION: RESUME AND APPLICATION MATERIALS NOTES

Canonical resume file: Yashwanth_Resume_v4_FINAL.docx.

Format note: LaTeX-generated PDF resumes were identified as problematic for ATS (Applicant Tracking System) parsers such as Workday, iCIMS, and Taleo, which is why the canonical resume is maintained as a .docx file instead.

Tailoring history: The resume has been tailored for multiple types of roles, including healthcare-sector roles, a Wingstop application (which scored 93/100 on an ATS compatibility check), and SRE (Site Reliability Engineering)-adjacent roles.

GitHub: A full audit of 18 repositories assessed Yashwanth's overall level as beginner-to-intermediate at the time, identified Aura Duo as the strongest differentiator, and flagged tutorial-style repos for privatizing. Professional READMEs were written for four of the strongest repositories. The GitHub profile README was later enhanced with a deep learning section and Aura Duo was promoted to a more prominent position.

SECTION: PERSONAL CONTEXT

Location: Based in Austin, Texas.

Vehicle: Drives a 2019 Honda Civic, and drives frequently.

Gaming: Owns a PlayStation 5 (PS5) and has an interest in gaming.

Fitness: Has been working on fitness goals using a structured gym routine.

Device setup: MacBook M1, Philips P27P1 external monitor, iPhone 16 Pro, VS Code as primary editor, Python 3.12 in a virtual environment.

SECTION: WORKING STYLE AND PREFERENCES (FOR ASSISTANT BEHAVIOR, NOT GENERAL FACTS ABOUT YASHWANTH)

Yashwanth prefers concise, human-like explanations grounded in practical data science and machine learning examples rather than academic framing.

He communicates directly and briefly, dislikes sugarcoating, and expects honest assessments even when the assessment is unfavorable.

He prefers code first, explanation after, with minimal filler.

He prefers that an assistant trust his stated workflow (for example, using draft placeholders with explicit intent to replace them before submission) rather than repeatedly re-checking or gatekeeping his process once he has stated his intent.

Other past academic or coursework-adjacent interests noted: Enterprise Systems / ERP / databases, data warehouse design, and AI-powered car diagnostics — these may reflect coursework exposure

rather than active personal projects.

END OF DOCUMENT